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A pilot rock magnetic and ore microscopy study of xenolith-bearing young basaltic rocks from the Camargo cinder cone field, Chihuahua, Northern Mexico

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Abstract

Initial results of a rock magnetic, paleomagnetic, microscopic study of the Camargo volcanic field from Chihuahua (northern Mexico) are reported. The field is formed by more than 300 cinder cones and extends over an area of 200,000 km^2 Mantle and lower crust xenoliths in the alkaline basaltic rocks of La Olivina, in the northern sector of the field, suggest rapid magma ascent rates. This study concentrates on the rock magnetic properties of the cinder cone field. Remanences are carried in most cases by Ti-poor titanomagnetites and magnetites as a result of the oxy-exsolution of original titanomagnetite during initial flow cooling. For a few units, remanence seems to be carried by (titano)hematites, as shown by unblocking temperature spectra and secondary titano(maghemite). Judging from the ratios of hysteresis parameters, all samples fall in the pseudo-single domain state region. Most units (13) show reverse polarity magnetizations, and only 2 show normal polarity. Radiometric dating gives the field's age range as between 2 and 1.7 Ma, which, together with the magnetostratigraphic results, suggests a relatively short emplacement time for the volcanic field. The mean preliminary paleomagnetic direction obtained for the Camargo volcanic field is $I = 30.6°$, $D = 341.3°$, $k = 19$, and $\alpha_{95} = 9.8°$, which deviates counterclockwise from the expected direction estimated from the North American apparent polar wander path. In turn, this suggests a vertical axis tectonic counterclockwise rotation of approximately 20° relative to stable North America.
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Keywords: Paleomagnetism; Northern Mexico; Pliocene; Camargo; Volcanic field

1. Introduction

The major tectonic volcanic province of western Mexico is the Sierra Madre Occidental (SMO), which represents the world's largest silicic province. It developed between Eocene and Miocene time over largely unextended crust that was surrounded by the extended crust of the Gulf Extensional and Basin and Range provinces (Henry and Aranda-Gomez, 2000). The area is crossed by a major proposed lithospheric boundary, represented by the Mojave-Sonora megashear, which produced large left-lateral displacement related to development of the Gulf of Mexico during the mid-Jurassic. The megashear also may have been active during the Tertiary. The widespread tectonic deformation and variety of styles—from continental-wide compression and extension to crustal and lithospheric thickening and thinning to tectonic uplift to lithospheric delamination to continental rifting—have initiated a new focus for paleomagnetic and tectonic studies. However, relatively few paleomagnetic studies have been conducted in northwestern Mexico (e.g. Urrutia-Fucugauchi, 1981; Goguitchaichvili et al., 2001).

Herein, we report the results of a rock magnetic study of the Camargo volcanic field (CVF), a compositional end member of the volcanic fields in the region (Fig. 1). The study's other objectives are to investigate any spatial–temporal migration of volcanism and estimate the duration of activity. The CVF has long attracted interest because of

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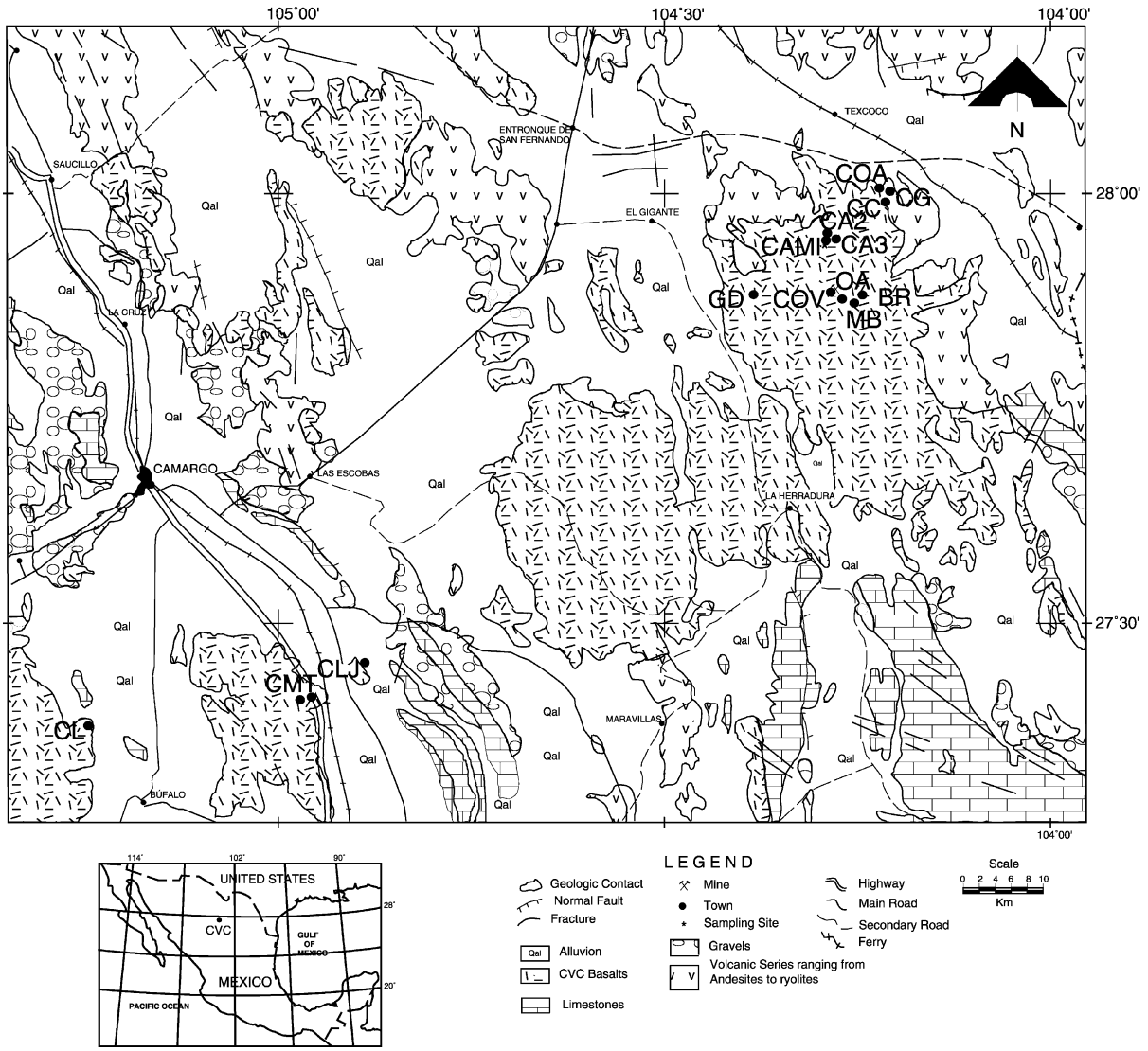


Fig. 1. Schematic map of Camargo volcanic field showing the location of the paleomagnetic sampling sites.

its occurrence of abundant lower crust and mantle xenoliths (Nimz et al., 1986, 1993). Models for cinder cone volcanism and the occurrence of xenoliths point to short magma ascent times without the development of mid-crust chambers. The xenoliths are present in cinder cones and lavas, which document rapid magma ascent rates through the crust. For example, the La Olivina piroxenite xenoliths occur in young 2 Ma basanites in the basaltic field. In contrast, silicic suites associated with caldera or stratovolcano activity are evolved and have been studied in previous paleomagnetic investigations (Urrutia-Fucugauchi, 1981; Goguitchaichvili et al., 2001). The young, well-exposed, fresh basaltic lavas enable investigation of the variations across the CVF, as well as comparisons of the paleomagnetic and rock magnetic properties of cinder cones in the central and adjacent sectors of the field with compositionally different volcanic suites.

2. Volcanism in northeastern Mexico and the CVF

Evolution of the western North American continental margin has been characterized by intense and widespread magmatic activity. Extensive areas in the southwestern United States and northwestern Mexico are covered by thick sequences of volcanic rocks whose characteristics and compositions are intimately related to plate tectonic and lithospheric/crustal evolution since the Mesozoic. Magmatic products can be found from the margin to more than 1200 km inland. Magmatism and tectonism have evolved as a result of large-scale plate interactions that result from long-term plate convergence, collision and subduction of a ridge system, and transform fault development. Crustal structure and magma compositions have evolved with regional changes from dominantly compressional to extensional tectonism. These processes have produced

the volcanic fields of the SMO, Trans-Pecos, and Southern Cordilleran Basaltic Andesitic suite, as well as the tectonic provinces of the Basin and Range, Rio Grande rift, and Gulf of California extensional province (Cameron et al., 1989; Henry and Aranda-Gomez, 2000).

These rich, diverse magmatic suites and variety of tectonic environments have long attracted the interest of petrologists, geochemists, and structural geologists, who have contributed data and tectonic-magmatic models. Paleomagnetic results indicate vertical axis tectonic rotations that have affected the volcanic fields (Goguitchaichvili et al., 2001). The vertical axis rotations could be related to regional tectonic stress fields or local processes of extension and lateral faulting. The diversity of tectonic and volcanic features provides an opportunity to investigate distinct aspects of the evolution of the region. In particular, we are interested in documenting the paleomagnetic record and rock magnetic properties of volcanic fields and investigating their temporal and geographic characteristics.

The CVF is located in the central eastern part of Chihuahua (Fig. 1), within the Basin and Range tectonomagmatic province with the following N-NW-trending mountain ranges: Sierra Agua de Mayo, Sierra Agua de Chile, Toba del Diablo, and Sierras Honorato and Guadalupe. The field has an approximate extension of 2000 km^2 and is formed by several alkaline basaltic plateaus and ca. 300 volcanic edifices (Rojas-Beltran, 1995). The field contains several localities with xenoliths and fills a broad basin east of Camargo. The xenoliths are mainly pyroxenites with isotopic ages ranging from 20 to 0 Ma and reflect high temperatures of granulite facies, common in the deep crust of Mexico in regions of Pliocene–Pleistocene volcanism (Rudnick and Cameron, 1991). The mountain ranges in the vicinity of La Olivina, which is the main xenolith locality (Nimz et al., 1986), are composed of mid-Cenozoic intermediate to silicic ignimbrites and lava flows (Cameron and Cameron, 1985). The mid-Cenozoic sequence is part of a voluminous, continental arc-related, magmatic event widespread throughout southwestern North America.

The CVF has been studied widely, with reports on geology, structural and economic geology, chemistry, isotopes, and petrology (Ugalde, 1956; Valencia, 1973; Muñoz-Saldua, 1975; Nimz et al., 1986; Cameron et al., 1989; Rudnick and Cameron, 1991; Nimz et al., 1993; Smith, 1993; Noyola-Medrano, 1995; Rojas-Beltran, 1995).

3. Paleomagnetic sampling

Paleomagnetic sampling was concentrated in the central northern sector of the field, which includes the xenolith-bearing lavas and pyroclastic deposits (Fig. 1). Additional sampling localities were selected in areas to the southeast to investigate any lateral migration of activity and develop a magnetostratigraphy for the CVF (Fig. 1).

In total, 126 oriented samples from 15 cooling units were collected. The outcrops extend laterally over a few meters, and in these cases, we generally drilled 8–11 cores per unit. At a few small outcrops, only 6 cores were collected. The samples were distributed throughout each flow, both horizontally and vertically, to minimize the effects of block tilting and lightning. In general, samples were obtained at the bottom of the flows in hopes of collecting samples with the finest grains of material. However, in a few cases, samples were obtained from the surface of outcrops (top of plateau), making the required in situ condition for paleomagnetic sampling uncertain. Cores were obtained with a gasoline-powered portable drill and oriented in most cases with a magnetic compass (after verifying that the compass was not affected by the remanent magnetization of the outcrop). Our field observations showed that all sampled lava flows were horizontal. No tilt correction was applied to any of the sites.

4. Rock magnetic properties

To identify the magnetic carriers responsible for the remanent magnetization, obtain information about their paleomagnetic stability, and assess the suitability of the studied samples for future paleointensity studies, several rock magnetic experiments were carried out (Figs. 2–6, Table 1). These experiments included (1) continuous susceptibility measurements at low temperatures (from −190 °C to room temperature), (2) continuous susceptibility measurements at high temperatures (from room temperature to 600 °C), (3) hysteresis measurements, and (4) microscopic observations under reflected light.

4.1. Low and high temperature continuous susceptibility curves (k–T curves)

The low temperature variation of susceptibility (from −190 °C to room temperature) was recorded using a Highmore susceptibility bridge equipped with a furnace in a Mexico City laboratory (one sample per unit).

There remains a great deficit in rock magnetic literature about the interpretation of low temperature continuous susceptibility curves. Two watershed papers (Senanayake and McElhinny, 1981; Radhakrishnamurty et al., 1982) yield contradictory results. More recent works (Özdemir et al., 1993; Dunlop and Özdemir, 1997) show only results from chemically well-identified synthetic magnetites, and only saturation isothermal remanent magnetization behavior is analyzed. Moskowitz et al. (1998) study is probably the most representative, because it provides low temperature k–T curves for natural volcanic samples that contain TM55, TM35, and TM10.

In our case, three different types of behavior were observed during experimentation (Fig. 2). For the first type, the curve shows a well-defined inflection point (Fig. 2(a)),

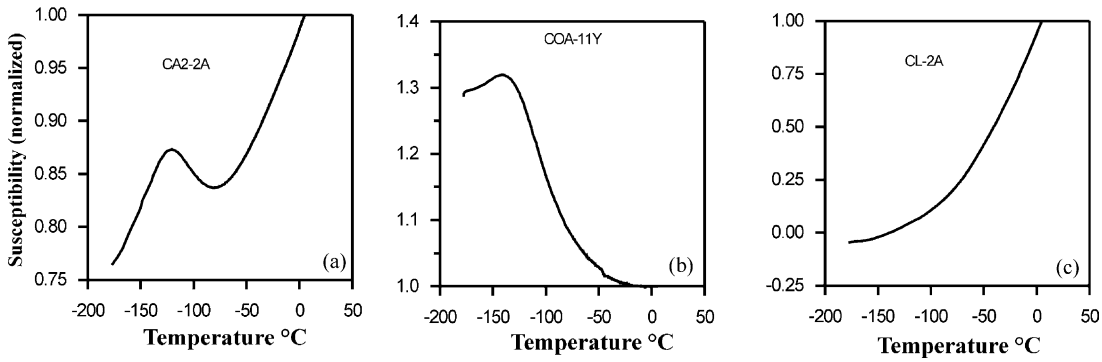


Fig. 2. Representative low temperature versus susceptibility curves.

which probably indicates a Verwey transition, characteristic of almost pure magnetite. Some superparamagnetic grains may also contribute to this behavior. The second type (Fig. 2(b)) is characterized by a monotonic decrease from −185 °C to room temperature that yields a poorly defined inflection point at approximately −145 °C. This behavior seems typical for Ti-rich titanomagnetite. The third type provides an almost exponential increase (Fig. 2(c)) without any inflection points, which points to titanomagnetites with reasonably high titanium content (TM35 after Moskowitz et al. (1998)).

For the high temperature susceptibility measurements, one sample from each site was heated to 650 °C at a heating rate of 20 °C/min and then cooled at the same rate. The correct interpretation of these curves is delicate, because the measurements were carried out under air. The Curie temperature was determined using Prévot et al. (1983) method. More than half the curves indicate the presence of a single magnetic/ferrimagnetic phase with a Curie point compatible to relatively low Ti titanomagnetite (Fig. 3(a) and (b)). However, the cooling and heating curves are not perfectly reversible, which may be due to oxidation during heating in air or experimental artifacts.

In other cases, $k–T$ curves yield two different thermomagnetic phases during heating (Fig. 3(c)). The lower Curie point ranges between 380 and 420 °C, whereas the highest is approximately 570 °C. The cooling curve shows only a single phase, with a Curie temperature close to that of magnetite. Such irreversible $k–T$ curves can be explained by titanomaghemite, which probably transformed into magnetite (Readman and O'Reilly, 1972; Ozdemir, 1987) during heating.

Finally, note that a correct susceptibility versus temperature curve could not be obtained for some sites because of the low initial signal of susceptibility.

4.2. Hysteresis measurements

Hysteresis measurements at room temperature were performed on all studied samples using the AGFM 'Micromag' apparatus in fields up to 1 T. The saturation remanent magnetization (J_{rs}), saturation magnetization (J_s), and coercive force (H_c) were calculated after correction for the paramagnetic contribution. The coercivity of remanence (H_{cr}) was determined by applying a progressively increasing backfield after saturation. Some typical hysteresis plots appear in Fig. 4. Slightly wasp-waisted behavior, which probably reflects the presence of ferrimagnetic phases with different coercivities (Tauxe et al., 1996), was obtained for one case (Fig. 4, sample CC-7Z). Alternatively, this behavior may be due to the presence of single domain (SD) and superparamagnetic grains (Goguitchaichvili and

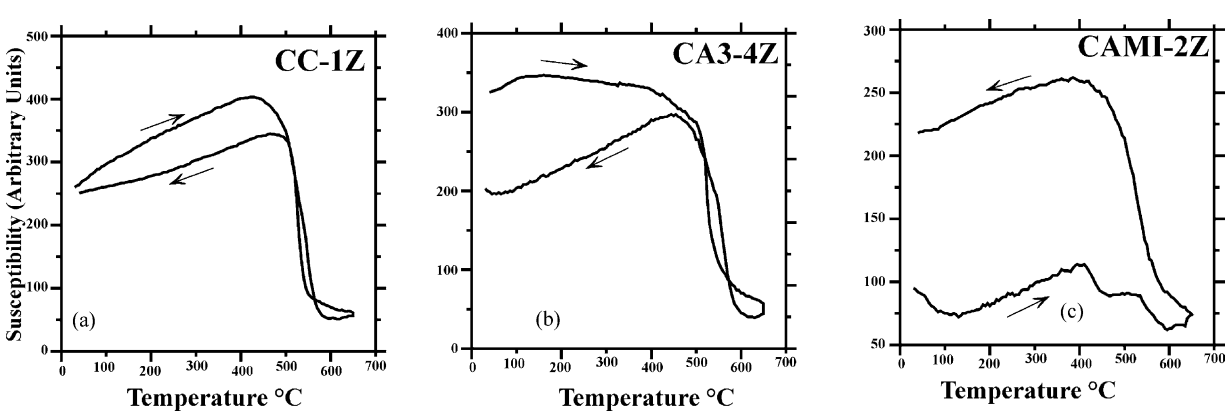


Fig. 3. Representative high temperature versus susceptibility curves. The arrows indicate heating and cooling curves.

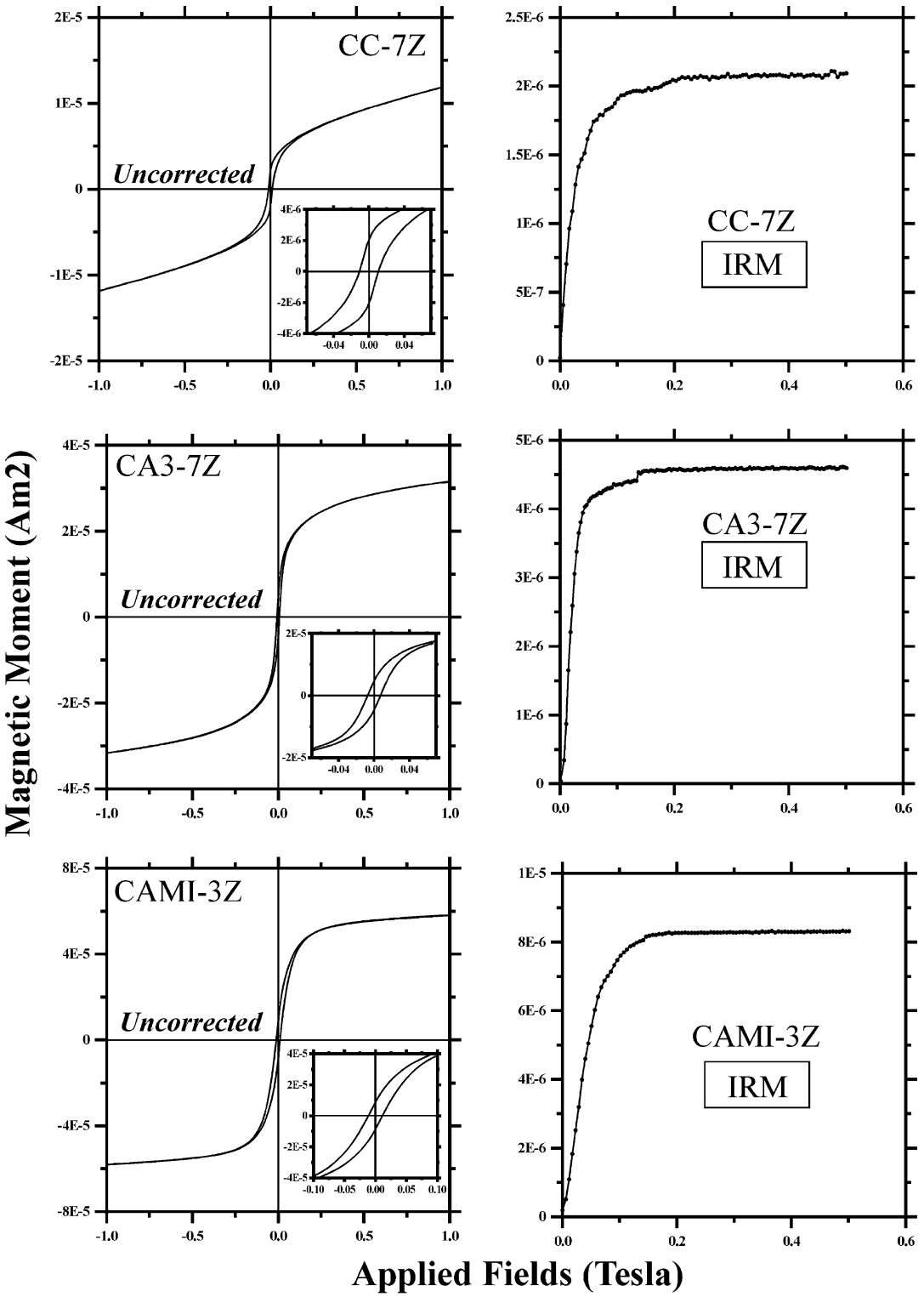


Fig. 4. Examples of hysteresis loops (uncorrected) of small chip samples.

Prévot, 2000). Other than this single case, the curves are symmetrical and similar. Isothermal remanent magnetization acquisition curves also were very similar for all samples. Saturation was reached in moderate fields (Fig. 4, right) on the order of 100–150 mT, which indicates that a spinel phase is the main remanence carrier. According to the ratios of the hysteresis parameters (Fig. 5), almost all samples fall in the pseudo-single domain (PSD) grain size region. As recently shown by Goguitchaichvili et al. (2001), domain state estimation using room temperature hysteresis

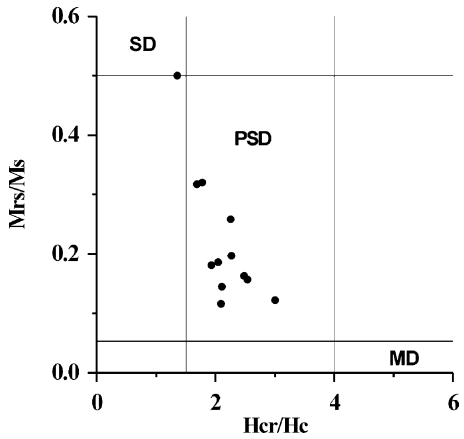


Fig. 5. Day diagram showing the possible magnetic domain type for our samples.

parameters for the plot of the magnetization ratio versus the coercivity ratio has no resolution for most natural rocks. In addition, PSD behavior on a Day plot may indicate a mixture of multidomain (MD) and a significant amount of SD grains.

4.3. Microscopic observations

Optical observations under transmitted and reflected light were made for one to two samples from each site. In general, the studied samples are classified as alkaline, hiperstene, and olivine basalts that present mainly two generation of Fe–Ti oxides. The larger crystals (more than 30 μm), which can be observed easily, are scarce and usually show exsolution textures (trellis and sandwich). The small (euhedral) crystals (3 μm and less) are abundant in all studied samples (Fig. 6(a)). There is also a significant quantity of intermediate crystals between 3 and 30 μm. Magnetic grains are irregularly distributed in the gray, sometimes reddish, matrix, which is mainly composed of plagioclase, clay, and oxy-hydroxides of Fe. Occasionally, plagioclase contains magnetic minerals. These intergrowths typically develop at temperatures greater than 600 °C, and consequently, the natural remanent magnetization carried on these samples should be thermoremanent magnetization (TRM). We note, however, that the magnetic grains are often too small and that the petrological/optical observations of

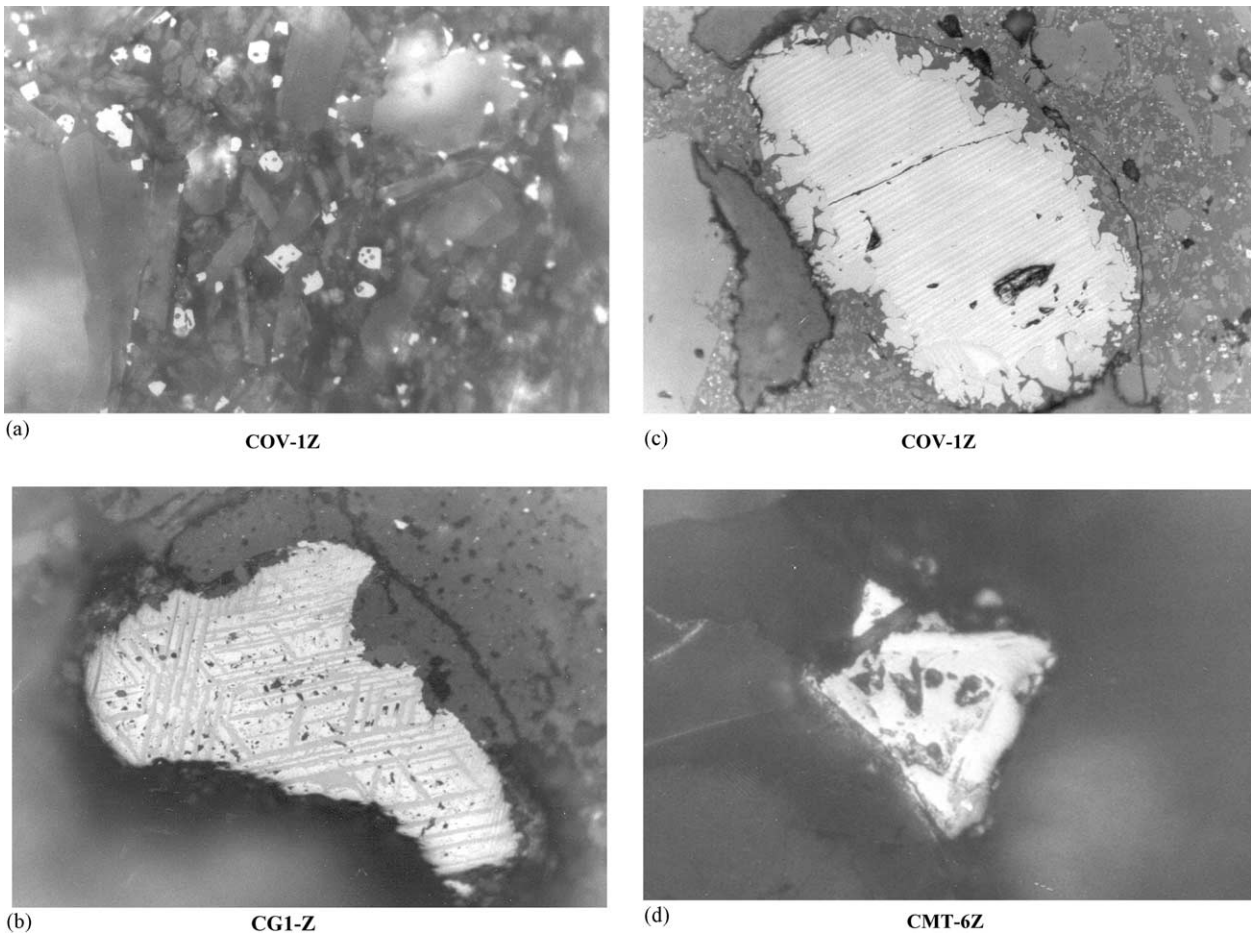


Fig. 6. Representative microphotographs of the opaque minerals. Subhedral to euhedral magnetite crystals (stage C1; Haggerty, 1976) disseminated in the whole sample are shown in (a) (1 mm = 1.38:m). Sample CG-1z (b) shows a titanohematite with ilmenite exsolution lamellas; incipient rutile is also detected (1 mm = 3.51:m), (stage C5; Haggerty, 1976). (c) shows some typical ilmenite reactions of low temperature oxidation. Sample CMT-6Z (d) presents a subhedral grain of original titanomagnetite with satellite skeletal extensions that may consist of pseudobrookite plus hematite. None of the original magnetite is present.

Table 1
Rock magnetic results for the Camargo volcanic field

Sample	M_r (μA m^2)	M_s (μA m^2)	M_r/M_s	H_c (mT)	H_{cr} (mT)	H_{cr}/H_c	K ($\times 10^{-5}$) SI	k/M_s (mA m^2)	W (mg)	J_r (A/m)	Magnetic domain	$J_i = kH$ (A/m)	Q
CMT-10Z	2.02	7.85	0.258	24.4	55	1.69	411	52.36	29	5243	PSD	15.63	35.54
CMT-15Z	8.04	25.4	0.317	29.6	49.9	1.69	828	32.60	36	21954	PSD	31.50	69.69
CG-5Z	529*	4.34	0.122	10.7	32.1	2.25	223	51.38	31	589	PSD	8.48	6.94
COA-3Z	5.10	26.00	0.197	23.9	54.2	2.27	1062	40.85	22	941.63	PSD	40.40	2.33
CL-2Z	3.56	30.8	0.116	5.54	11.6	2.09	1558	50.58	30	12685.7	PSD	59.27	21.40
CC-7Z	2.03	6.35	0.320	11.1	19.7	1.78	456	71.81	29	39970	PSD	17.35	23.03
CLJ-7Z	2.90	18.5	0.157	10.3	26.1	2.53	801	43.30	30	25741.1	PSD	30.47	84.48
COV-5Z	7.82	41.9	0.186	17.9	36.6	2.05	1552	37.04	30	3226.88	PSD	59.04	54.66
COV-8Z	9.09	18.2	0.500	73.8	100	1.35	840	46.15	29	17956.1	PSD	31.95	56.20
CAMI-3Z	8.66	53.1	0.163	11.3	28.0	2.48	2794	52.62	29	16286.1	PSD	106.28	15.32
CA2-3Z	5.37	37.1	0.145	6.93	14.6	2.11	2687	72.43	27	7701.5	PSD	102.21	7.53
CA3-7Z	4.63	25.7	0.181	7.26	14.0	1.93	1731	67.35	29	2787.31	PSD	65.85	4.23

M_r, remanent magnetization; M_s, saturation magnetization; H_c, coercive force; H_{cr}, coercivity of remanence; k, low field magnetic susceptibility; W, weight; J_i, induced magnetization; J_r, natural remanent magnetization; and Q, Koenigsberger ratio.

larger ferrimagnetic minerals may not correlate completely with the magnetic behavior of the whole rock.

In most cases, we were able to detect evidence of a pseudomorphed grain of titanomagnetite plus ilmenite. The assemblage is rutile plus hematite, and the original [111] fabric is maintained by moderated concentrations of rutile in areas that originally were occupied by ilmenite and by higher concentrations of hematite in areas that originally were titanomagnetite (Fig. 6(b)) (Haggerty, 1976). In other cases, the crystal is divided by several thin, gray, subparallel exsolution lamellae (Fig. 6(c)). The rest of the crystal presents high reflectivity and remarkable anisotropy, characteristic of (titano)hematite. The host of this grain is ilmenite, and the oriented lamellae are magnetites that show partial to complete decomposition to titanomaghemite. The core abounds with lamellae, whereas the margins are free of them. The lamellae result from subsolidus reduction and subsequent low temperature oxidation (Haggerty, 1976).

The opaque minerals observed for sample CMT-6Z (Fig. 6(d)) correspond to spinels without any apparent exsolution. Neither anisotropy nor pleochroism was observed for those crystals. Near the crystal's borders and fractures appears more reflective, which indicates some alteration. These areas seem to correspond with highly oxidized titanomaghemite, in agreement with thermomagnetic curves, because the presence of titanomaghemite was evidenced for some units. The magnetization carried by these grains is probably chemical remanence (CRM) (Goguitchaichvili et al., 2000). The crucial question is whether the CRM is acquired parallel to the primary TRM. Most experimental studies indicate that chemical magnetization due to maghemitization of magnetite (Johnson and Merrill, 1974; Heider and Dunlop, 1987) or titanomagnetite (Ozdemir and Dunlop, 1985; Nishitani and Kono, 1989) is recorded in the same direction as the original TRM. However, this may be not true for natural volcanic samples, which contain (titano)hematite as a hydrothermal product (Walderhaug et al., 1991; Walderhaug, 1992).

5. Paleodirectional results

The remanent magnetization of six to twelve samples from each unit was measured with a JR-5A spinner magnetometer (sensitivity $\sim 10^{-9}$ Am^2). Measurements were recorded after stabilization of the magnetometer's remanence. Both alternating field (AF) demagnetization, using a laboratory-made AF demagnetizer, and stepwise thermal demagnetization of up to 575–675 °C, using a noninductive Schonstedt furnace, were carried out. During thermal demagnetization, the low field susceptibility at room temperature was measured after each step with a Bartington susceptibility meter.

In most of studied units, only one paleomagnetic component could be recognized (Fig. 7(a) and (b)). Very small secondary components were easily removed by applying 200 °C, which may indicate the presence of Ti-rich titanomagnetites. The greater part of remanent magnetization was removed at temperatures between 500 and 575 °C, which indicates that low Ti titanomagnetites are responsible for remanent magnetization. Nevertheless, an important fraction of higher unblocking temperatures can be observed for site CG (andesitic rhyolite, Fig. 6(a)), which points to (titano)hematite as a probable remanence carrier, in agreement with microscopic observations. Two magnetization components were detected for the COV and CAMI sites (Fig. 7(c) and (d)). For these sites, the direction of primary magnetization was obtained after applying a 30 mT alternating magnetic field (Fig. 7(c)).

A characteristic magnetization direction was determined using least squares (Kirschvink, 1980); four to nine points

were taken in the principal component analysis for this determination. The obtained directions were averaged by unit, and the statistical parameters were calculated assuming a Fisherian distribution. The average unit directions are rather precisely determined for 13 sites (Table 2(a), Fig. 8), which yielded α_{95} of less than 15. Sites CMT1 and CLJ are characterized by large dispersion. Moreover, they show relatively high intensity, scattered NRM directions, and a magnetic mineralogy dominated by secondary titanomaghemite. These samples were rejected for further paleomagnetic analysis.

6. Discussion and main results

The characteristics and relation of rock magnetic property, as well as the variations in crystal fractionation and crustal assimilation processes associated with a long residence time in shallow crustal magma chambers (which occur in calderas and stratovolcanoes), may present distinct features compared with mafic alkaline basalts that rapidly extruded in cinder cone volcanism in extended tectonic domains. However, characterization of the rock magnetic properties of these rocks has been poor. An interesting aspect of the development of the CVF is related to the duration of activity and the extrusion rates, which can be investigated through magnetic polarity stratigraphy and paleosecular variation of the magnetic field. Magnetostratigraphic results could be useful in investigating spatial–temporal patterns and the migration of activity across the field. The documentation of normal and reverse polarities and intermediate transitional magnetizations, when combined with radiometric, volcanic, and detailed geologic studies, may assist in understanding the volcanic stratigraphy and construction of the Camargo cinder cone field.

The characteristic paleomagnetic directions determined in this study are of primary origin for 13 sites. This is confirmed by the determination of normal and reversed polarities. In addition, thermomagnetic curves show that remanence is carried, in most cases, by Ti-poor titanomagnetite, a result of the oxy-exsolution of original titanomagnetite during the initial flow cooling. This in turn probably indicates a thermoremanent origin of

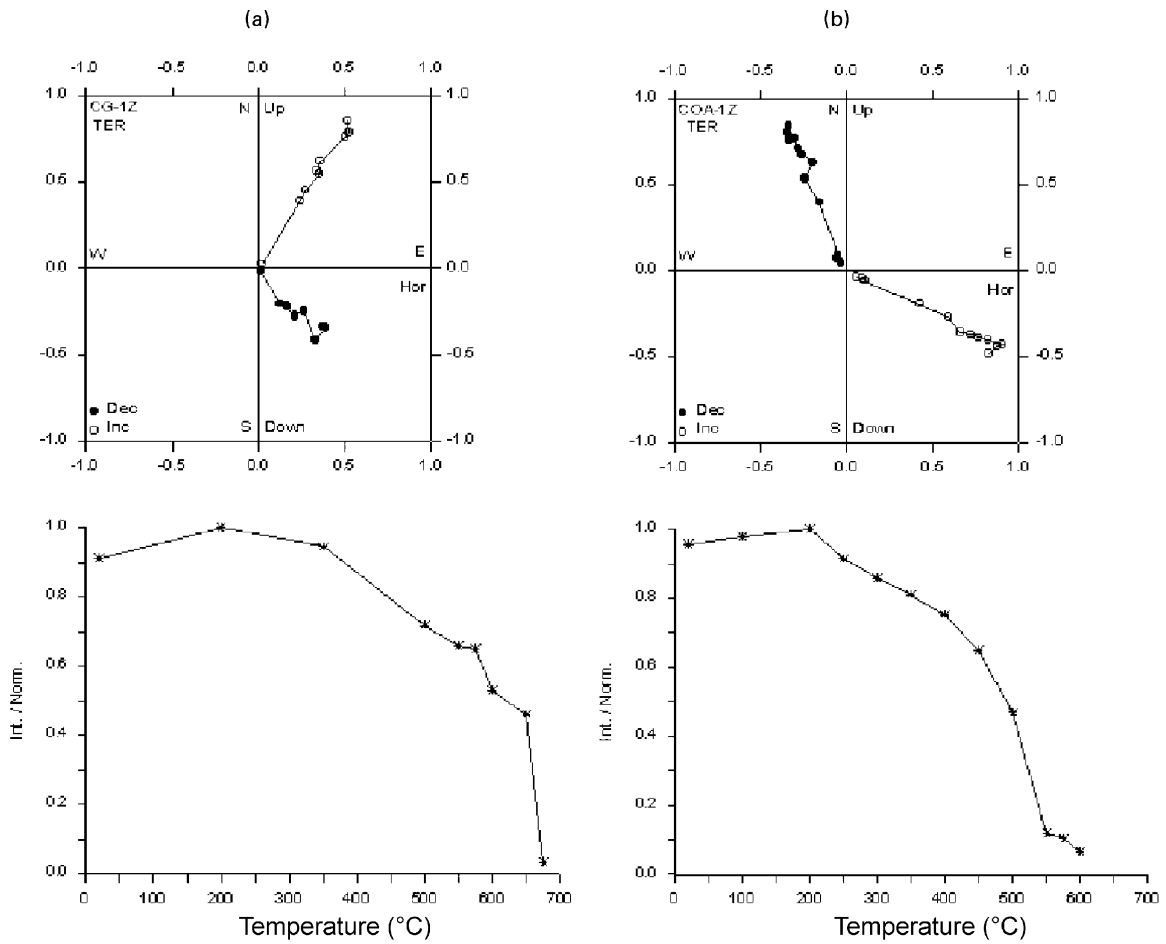


Fig. 7. Orthogonal vector plots of stepwise thermal or alternating field demagnetization (stratigraphic coordinates) of representative samples. The numbers refer to either temperatures in °C or peak alternating fields in mT. o, projections into the horizontal plane; x, projections into the vertical plane.

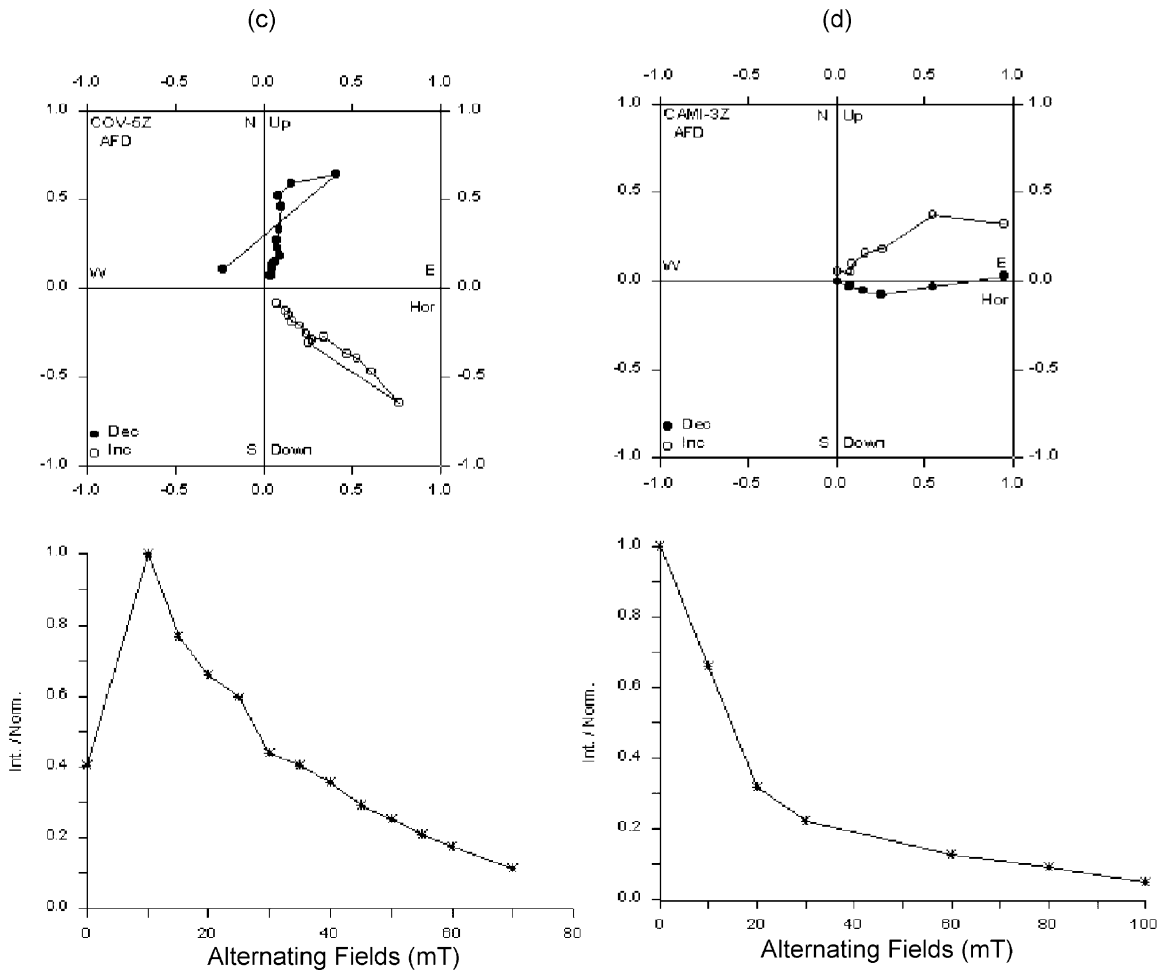


Fig. 7 (*continued*)

primary magnetization. Moreover, unblocking temperature spectra and relatively high coercivity point to a mixture of SD (dominant) and MD magnetic structure grains as responsible for remanent magnetization. Single component, linear demagnetization plots were observed in most cases.

Most of studied sites yield reverse polarity magnetization, but two are normally magnetized. We found no intermediate polarity magnetization. Because of the uncertainty regarding the age of the rock sampled, it is not possible to make a precise correlation with the geomagnetic polarity time scale (GPTS). More precise Ar^{39}–Ar^{40} radiometric ages are needed before these magnetozones can be correlated with the reference GPTS.

Our paleomagnetic data set has a relatively large between-site angular dispersion (Fig. 8, Table 2). The mean direction was calculated after discarding sites CLJ and CMT1. The mean paleodirection obtained is $I = 30.6$, $D = 341.3°$, $k = 19$, $\alpha_{95} = 9.8°$, and $N = 13$ (Table 2). The paleomagnetic declination is concordant with the recently determined Eocene–Oligocene paleoinclination (Fig. 8, Table 2(b)) for Chihuahua ($D = 337.1$, after Goguitchaichvili et al. (2002)) but differs from the expected Pliocene and Eocene–Oligocene magnetic declination (as derived from the reference poles for North America provided by Besse and Courtillot (1991)). This suggests a mean tectonic counterclockwise rotation of approximately 20°.

Although many detailed studies have been devoted to southern North America (Urrutia-Fucugauchi et al., 1981; Cameron et al., 1989; Nimz et al., 1986; Henry and Aranda-Gomez, 2000; Goguitchaichvili et al., 2001), there is no general consensus about the paleotectonic evolution of the area. Paleomagnetic directions are useful tools to estimate tectonic deformations. Previous paleomagnetic studies from Eocene–Oligocene volcanic rocks from northern Mexico (Urrutia-Fucugauchi, 1981; Goguitchaichvili et al., 2002) indicate a counterclockwise rotation (relative to stable North America). However, the time when deformation occurred and the rotation mechanism remained controversial issues, largely because no paleomagnetic results were available from the younger rocks. Our paleomagnetic data

Table 2
(a) Paleodirectional results from Chihuahua Oligocene volcanic rocks and (b) between-flow average paleodirections for the Eocene–Oligocene–Pliocene volcanic provinces from southern North America

(a) Site	*n/N*	*D*	*I*	*k*	α_{95}	POL	Plat	Plong
CMT	7/10	326.8	24.6	22.3	13.8	N	55.9	147.7
CMT1*	4/7	186.5	−51.8	7.6	24.1	R?	−82.2	120.1
CL	4/6	148.1	−33.8	32.6	11.8	R	−59.5	336.1
CLJ*	3/7	122	−38.5	4.1	51.6	R?	−37.3	353.2
COA	11/12	342.2	33	79	5.2	N	71.3	140.4
CG	6/6	144.1	−54.3	157	5.4	R	−58.4	8.8
CC	6/8	166.9	−17.5	59.6	10.8	R	−68.1	292.4
CAMI	4/6	136	−23.5	48.6	8.6	R	−46.3	334.7
CA2	4/6	171.9	−6.4	28.3	14.4	R	−64.9	274.9
CA3	7/9	185.6	−18.2	12.3	11.2	R	−71.6	237.8
COV	10/11	174	−40.2	68.5	7.6	R	−83.2	209.8
DA	7/8	142.7	−45.9	111	5.8	R	−56.7	357.3
MB	5/8	169.5	−40.8	58	10.1	R	−80.2	335.2
GD	8/9	178.1	−25.3	169	4.3	R	−77.6	264.5
BR	9/9	160.2	−23.3	70	6.2	R	−67.1	313.9

(b) Locality	*N*	*D*	*I*	*k*	α_{95}	Age	Ref.
La Junta (Chih)	12	340.3	49.4	12	12.8	~40	Goguitchaichvilietal.(2002)
S. Antonio (Chih)	12	334.6	47.4	45	6.5	33–25	Urrutia-Fucugauchi (1981)
CVC (Chih)	13	341.3	30.6	19	9.8	2.0–1.7	This study

N, number of treated samples; *n*, number of samples used for calculation; *D*, declination; *I*, inclination; *k* and α_{95}, precision parameter and radius of confidence cone; Pol, magnetic polarity; Plat/Plong, latitude/longitude of VGP position; Age, K–Ar ages in My.

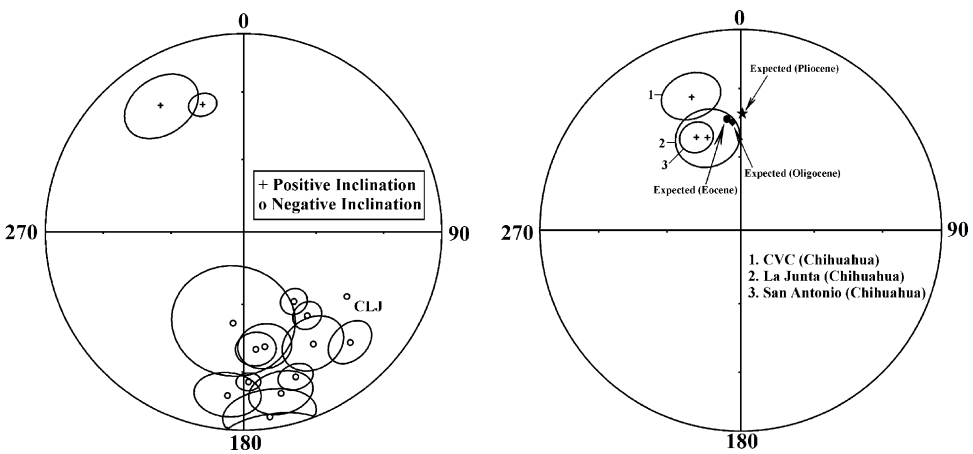


Fig. 8. (a) Equal area projections of the flow mean characteristic paleodirections for the Camargo volcanic field area (Chihuahua). Circles/crosses denote to the negative/positive inclination. (b) Equal area projection of the mean directions of Eocene–Oligocene–Pliocene volcanic units from southern North America, as indicated in Table 2.

yield the first evidence of counterclockwise rotations for Late Pliocene time, in contrast with previous studies, which attributed deformation to Laramide compression or Basin and Range extension in the Early Tertiary.

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References

Besse, J., Courtillot, V., 1991. Revised and synthetic apparent polar wander paths of the African Eurasian, North American and Indian Plates, and true polar wander since 200 Ma. J. Geophys. Res. 96, 4029–4050.

Cameron, K.L., Cameron, M., 1985. Rare earth element Sr/Sr and Nd/Nd composition of Cenozoic orogenic dacites from Baja California, northwestern Mexico, and adjacent west Texas: evidence for the predominance of a subcrustal component. Contrib. Mineral. Petrol. 91, 1–11.

Cameron, K.L., Nimz, G.J., Kuentz, D., Niemeyer, S., Gunn, S., 1989. The southern cordilleran basaltic andesitic suite, southern Chihuahua, México: a link between Tertiary continental arc and flood basalt magmatism in North America. J. Geophys. Res. 94, 7817–7840.

Dunlop, D., Özdemir, Ö., 1997. Rock Magnetism, Fundamentals and Frontiers, Cambridge University Press, Cambridge, 573pp.

Goguitchaichvili, A., Prévot, M., 2000. Magnetism of oriented single crystals of hemo-ilmenite showing self-reversal of thermoremanent magnetization. J. Geophys. Res. 105, 2761–2781.

Goguitchaichvili, A., Chauvin, P., Roperch, M., Prévot, M., Vergara, L., Moreno, H., 2000. Paleomagnetism of the Miocene Farelonnes formation in Chile. Geoph. J. Int. 140, 357–374.

Goguitchaichvili, A., Alva-Valdivia, L.M., Urrutia-Fucugauchi, J., Zesati, C., Caballero, C., 2001. Paleomagnetic and paleointensity study of Volcanic rocks from Chihuahua (Northern Mexico). Phys. Earth Planet. Int. 124, 223–236.

Haggerty, S.E., 1976. Oxidation of Opaque Mineral Oxides in Basalts, Oxide Minerals, vol. 3. Mineral Society of America, 300pp.

Heider, F., Dunlop, D., 1987. Two types of chemical remanent magnetization during oxidation of magnetite. Phys. Earth Planet. Int. 46, 24–45.

Henry, C.D., Aranda-Gomez, J.J., 2000. Plate interactions control middle-late Miocene, proto-Gulf and Basin and Range extension in the southern Basin and Range. Tectonophysics 318, 1–26.

Johnson, H.P., Merrill, R.T., 1974. Low-temperature oxidation of a single domain magnetite. J. Geophys. Res. 79, 5533–5534.

Kirschvink, J.L., 1980. The least-square line and plane and analysis of palaeomagnetic data. Geophys. J. R. Astron. Soc. 62, 699–718.

Moskowitz, B.M., Jackson, M., Kissel, C., 1998. Low-temperature magnetic behavior of titanomagnetites. Earth Planet. Sci. Lett. 157, 141–149.

Muñoz-Saldua, R., 1975. Estudio geologico de la procion occcidental de la Sierra de Encinillas, Camargo, Chihuahua. Unpublished Thesis. Escuela de Ingenieria UASLP, Mexico.

Nimz, G.J., Camereon, K.L., Cameron, M., Morris, S.L., 1986. The petrology of the lower crust and upper mantle beneath southeastern Chihuahua Mexico: a progress report. Geof. Int. 25, 85–116.

Nimz, G.J., Cameron, K.L., Niemeyer, S., 1993. The La Olivina pyroxenite suite and the isotopic compositions of mantle basalts parental to the mid-Cenozoic arc volcanism of northern Mexico. J. Geophys. Res. 98, 6489–6509.

Nishitani, T., Kono, M., 1989. Effect of low-temperature oxidation on the remanence properties of titanomagnetites. J. Geomag. Geoelectr. 41, 19–38.

Noyola-Medrano, M.C., 1995. Estudio comparativo de la geologia y morfologia de algunos conos cineriticos en los campos volcanicos de Camargo y de San Quintin, B.C. Unpublished Thesis. Facultad de Ingenieria, UASLP, Mexico.

Ozdemir, O., 1987. Inversion of titanomaghemites. Phys. Earth Planet. Int. 46, 184–196.

Ozdemir, O., Dunlop, D., 1985. An experimental study of chemical remanent magnetizations of synthetic moonodomain titanomaghemites with initial thermoremanent magnetizations. J. Geophys. Res. 90, 11513–11523.

Ozdemir, O., Dunlop, D., Moskowitz, B.M., 1993. The effect of oxidation on the Verwey transition in magnetite. Geophys. Res. Lett. 20, 1671–1674.

Prévot, M., Mankinen, E.A., Grommé, S., Lecaille, A., 1983. High paleointensity of the geomagnetic field from thermomagnetic studies on rift valley pillow basalts from the middle Atlantic ridge. J. Geophys. Res. 88, 2316–2326.

Radhakrishnamurty, C., Likhite, S.D., Deuthsch, E.R., Murthy, G.S., 1982. On the complex magnetic behavior of titanomagnetites. Phys. Earth Planet. Int. 30, 281–290.

Readman, P.W., O'Reilly, W., 1972. Magnetic properties of oxidized (cation-deficient) titanomagnetites (Fe, Ti,)O4. J. Geomag. Geoelec. 24, 69–90.

Rojas-Beltran, M.A., 1995. El sector nor-oriental del campo volcanico de Camargo, Chihuahua: geologia regional y petrologia de los xenolitos del manto, una comparacion con el campo volcanico de San Quintin, B.C. Unpublished Thesis. UASLP, Mexico.

Rudnick, R.L., Cameron, K.L., 1991. Age diversity of the deep crust in northern Mexico. Geology 19, 1197–1200.

Senanayake, W.E., McElhinny, M.W., 1981. Hysteresis and susceptibility characteristics of magnetite and tittanomagnetites: interpretation of results from basaltic rocks. Phys. Earth Planet. Int. 26, 47–55.

Smith, D.R., 1993. The Agua de Mayo mid-Cenozoic volcanic group and related xenoliths from La Olivina SE Chihuahua, Mexico. Unpublished Ms. Thesis. University of California (Saanta Cruz).

Tauxe, L., Mullender, T.A.T., Pick, T., 1996. Pot-bellies, wasp-waists and superparamagnetism in magnetic hysteresis. J. Geophys. Res. 95, 12337–12350.

Ugalde, H., 1956. Reconocimiento geologico del area de Camargo-Boquilla-Jimenez, Chihuahua. Unpublished Report, Petroleos Mexicanos.

Urrutia-Fucugauchi, J., 1981. Paleomagnetic evidence for tectonic rotation of northern Mexico and the continuity of the Cordilleran orogenic belt between Nevada and Chihuahua. Geology 9, 178–183.

Valencia, R.J., 1973. Reconocimiento geologico de las areas de La Boquilla. Unpublished Report, Petroleos Mexicanos.

Walderhaug, H., 1992. Directional properties of alteration CRM in basic igneous rocks. Geophys. J. Int. 111, 335–347.

Walderhaug, H., Torsvik, T.H., Lovlie, R., 1991. Experimental CRM production in a basaltic rock: evidence for stable, intermediate paleomagnetic directions. Geophys. J. Int. 105, 747–756.